

Portfolio managers could improve fund performance by analyzing appropriate holding sizes based on both alpha expectations and trading costs.

MI FACTOR SCORES

The preceding section showed how portfolio managers could back into the order size that could be transacted for a specified cost. But this often requires complicated numerical procedures to find the solution and this must be solved for each stock individually. This analyses can be done, but they are often quite time consuming.

The market impact factor score (MI Factor Score) is an alternative method to efficiently screen stocks based on trading costs. The MI Factor Score incorporates the I-Star market impact model, corresponding parameters, and stock specific trading characteristics (liquidity, volatility, and market price) to determine a trading cost score. The higher the score the more expensive the stock is to trade and the lower the score the less expensive the stock is to trade.

Manager's use the MI Factor Score to improve their stock screening process by filtering for the more expensive and difficult names to trade. In addition to being a direct calculation, this provides an improvement over methodologies that only consider liquidity or only consider volatility when filtering for trading costs.

The advantage to our MI factor score is that it does not require complicated or sophisticated numerical procedures, and it is a more accurate representation of the trading cost environment. The derivation of the MI factor score is as follows:

Derivation of the MI Factor Score for Shares

Step 1. Start with the I-Star model:

$$I^*(Shares) = a_1 \cdot \left(\frac{Shares}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \quad (11.11)$$

Step 2. Rearrange the terms in the expression by factoring out shares:

$$I^*(Shares) = \left\{ a_1 \cdot \left(\frac{1}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \right\} \cdot Shares^{a_2} \quad (11.12)$$